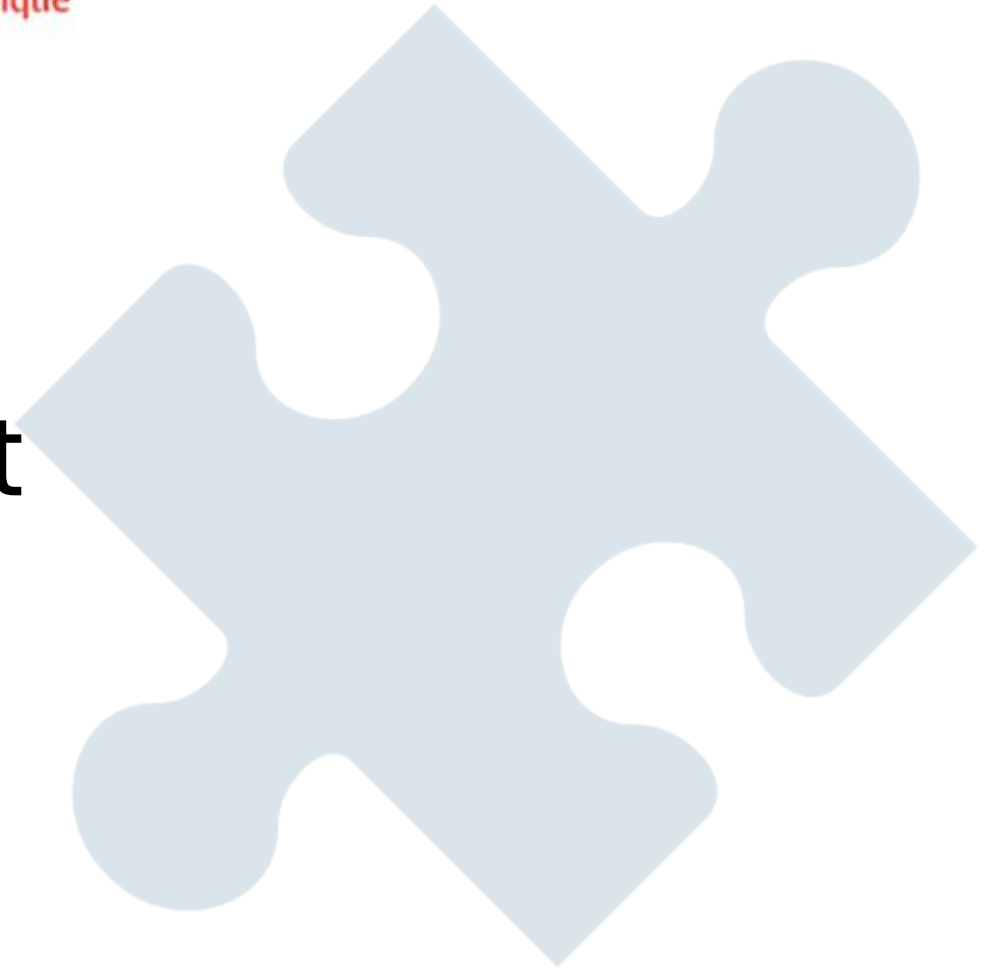


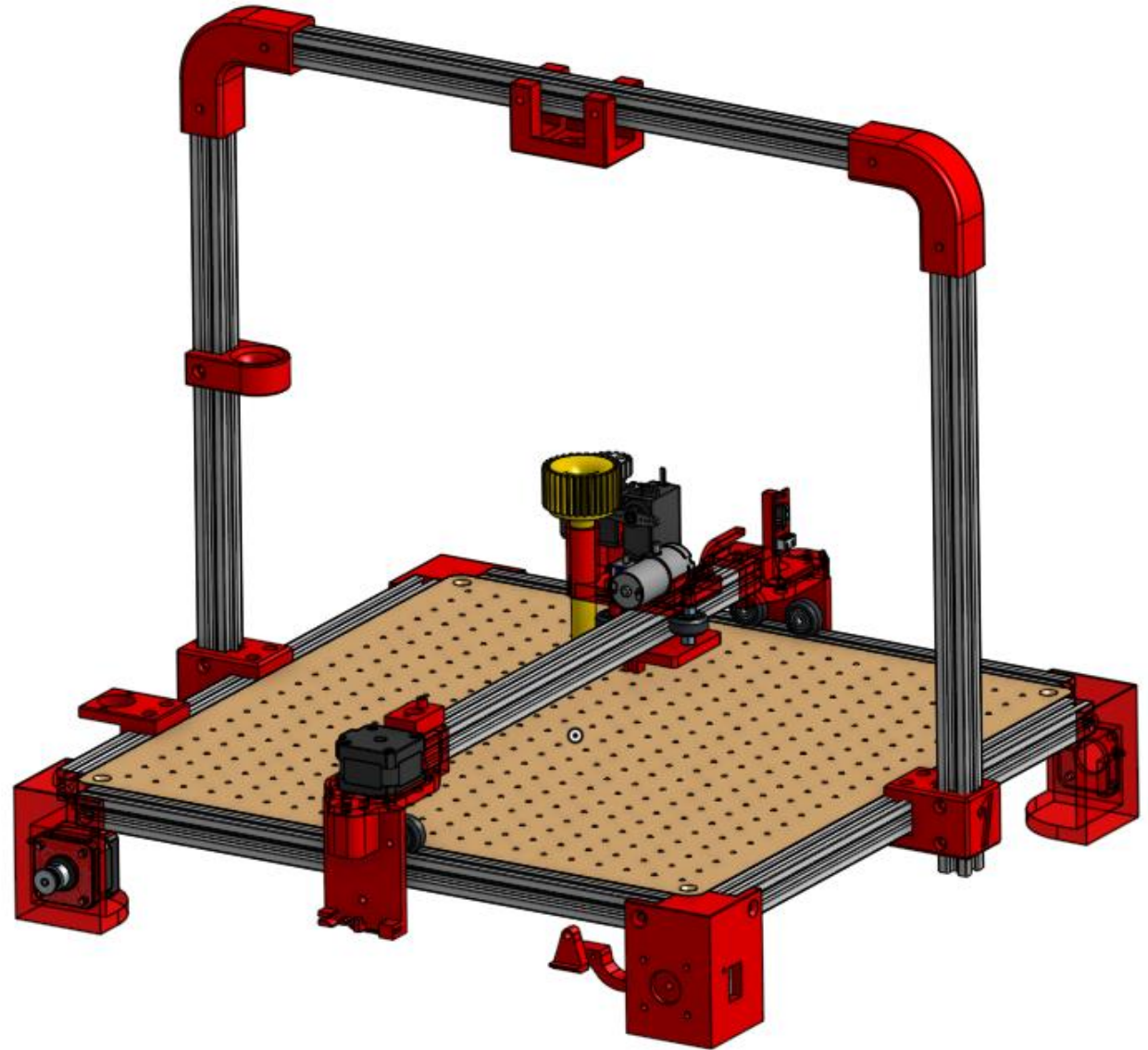


Projet I3 : Puzzle Bot



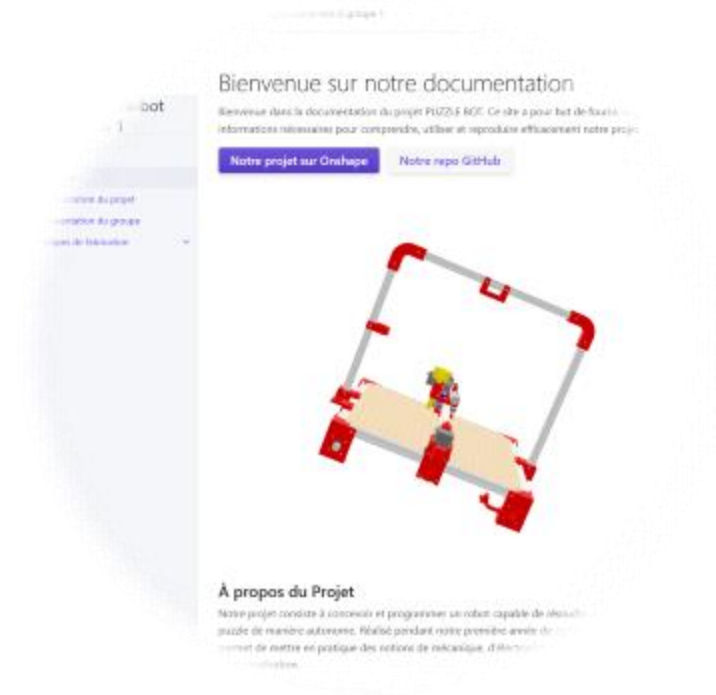
Introduction

- Contexte et objectifs
- Conception mécanique
- Electronique
- Pompe
- Caméra
- Programmation
- Conclusion



Contexte et objectifs

- Projet de cours
- Inspirations
- "Proof of Concept" (POC)
- Github (contexte, objectifs, cahier des charges)
- Répartition
- 50 heures



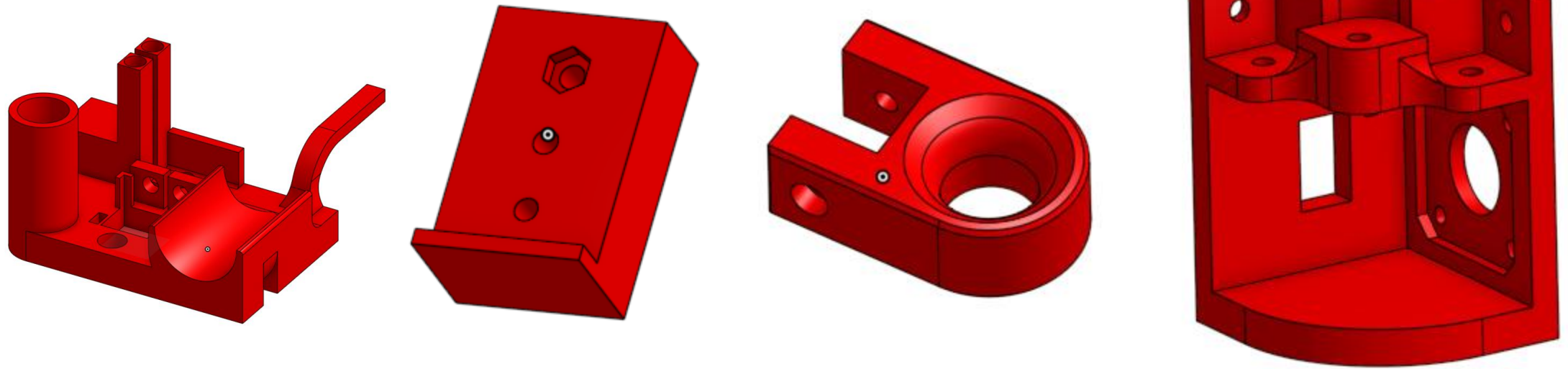
Conception
mécanique

Electronique /
Pompe

Caméra /
Assemblage

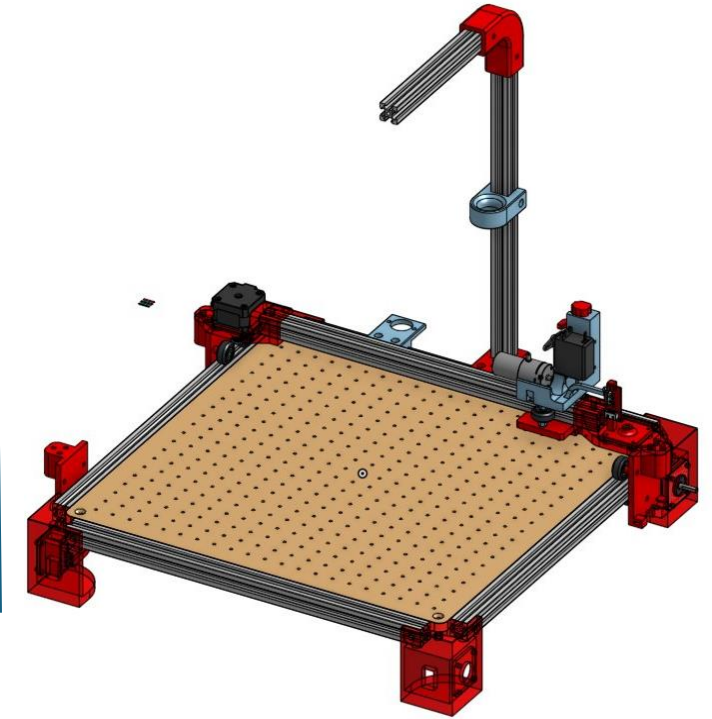
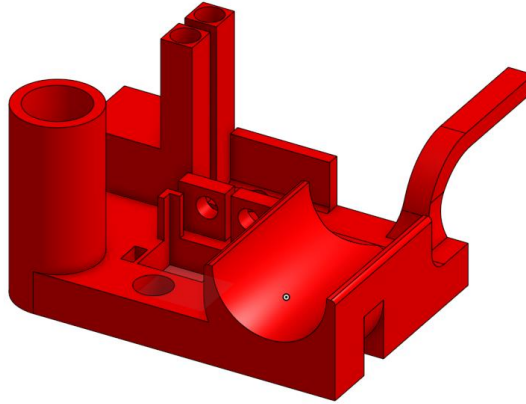
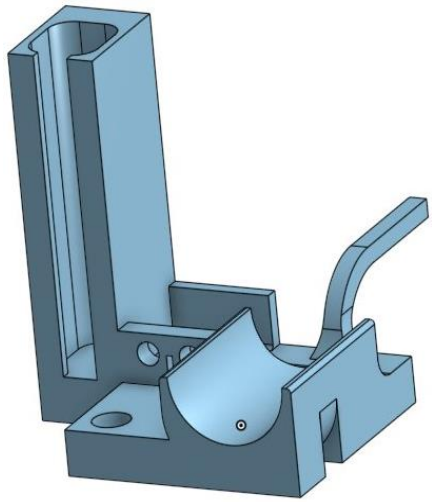
Première
version

Peaufinage



Conception mécanique

- CAO
- Nouveaux supports

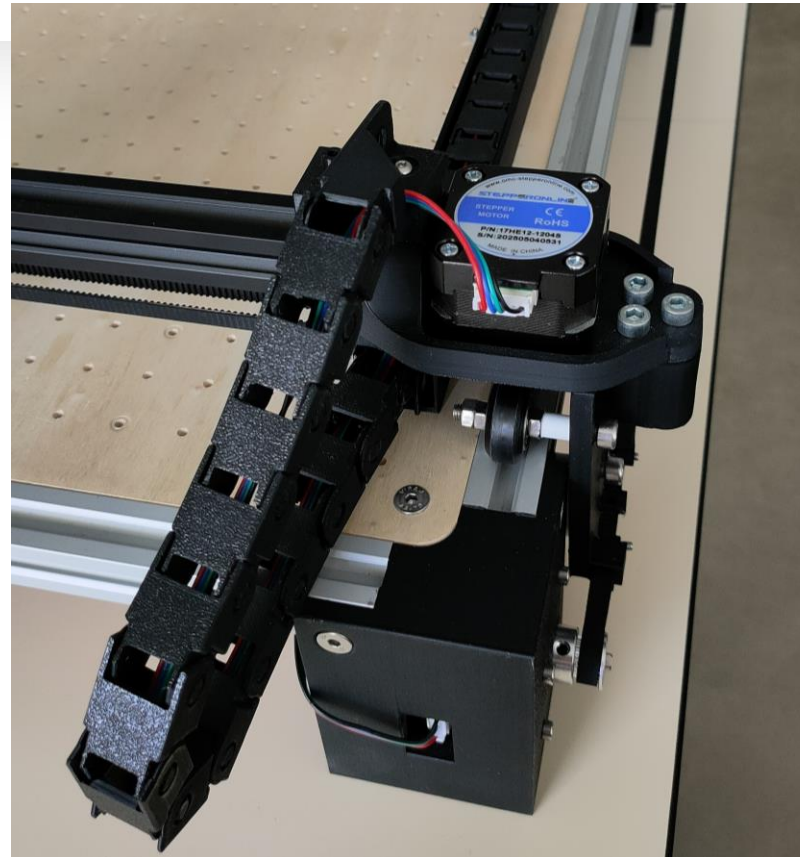
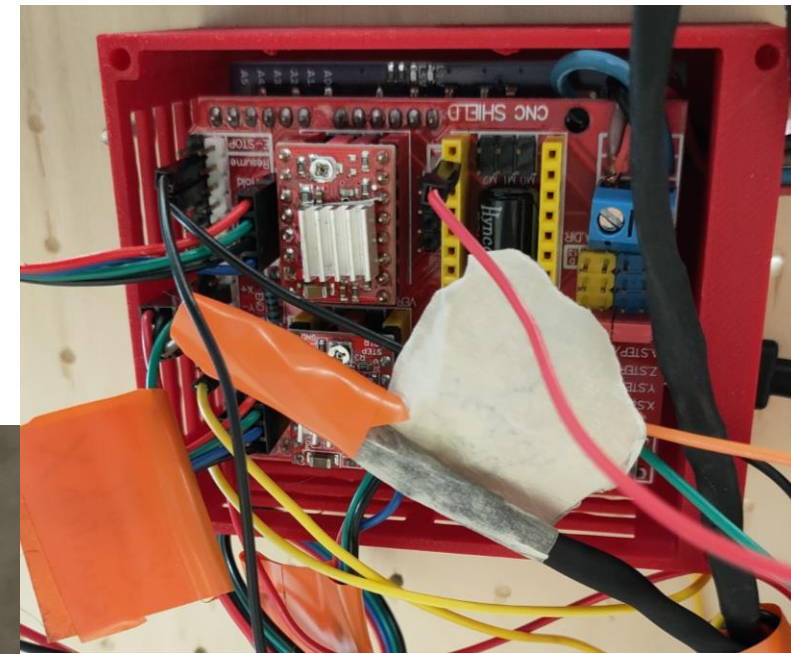


Conception mécanique

- Améliorations

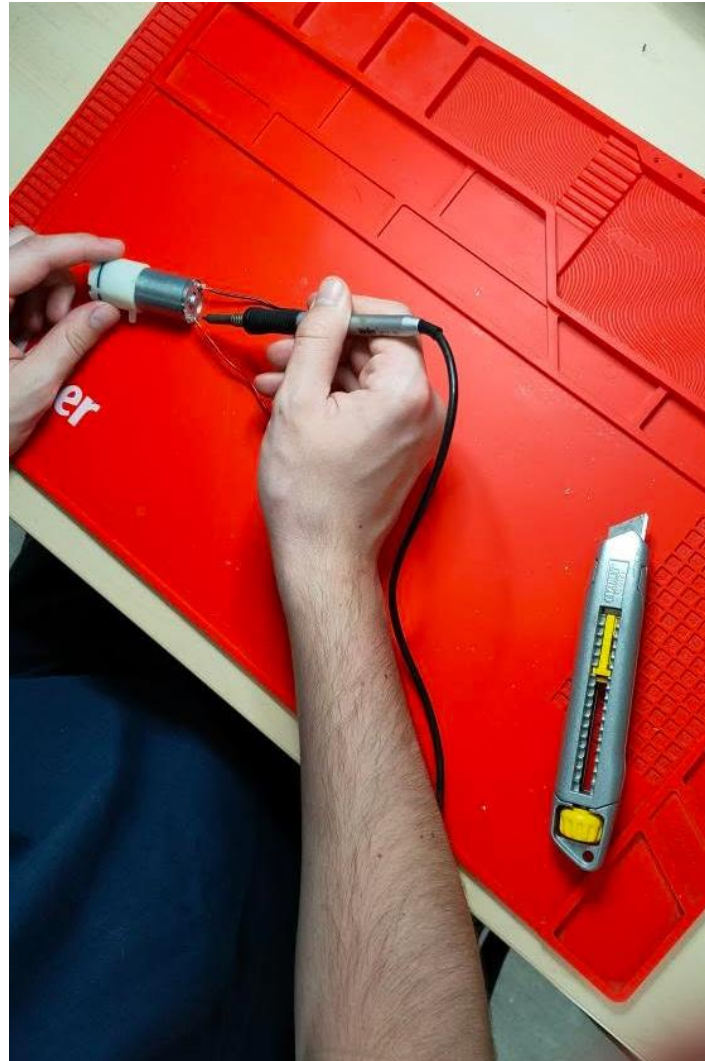
Electronique

- Shield
- Moteurs
- CNC
- Carte Arduino
- Réglage des drivers
- Essais concluants

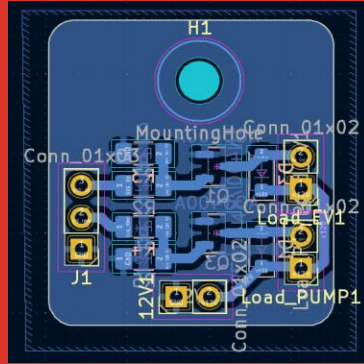


Pompe

- Délais avant d'avoir la pompe
 _ Etude des branchements
- Soudure des fils
- Test code
- Problème de voltage



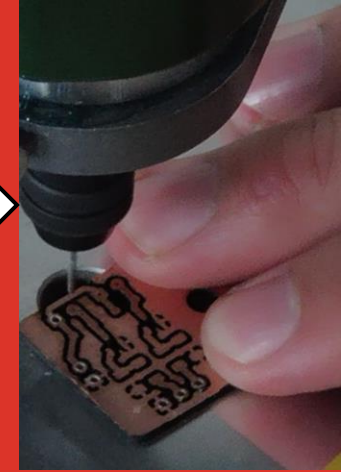
Pompe



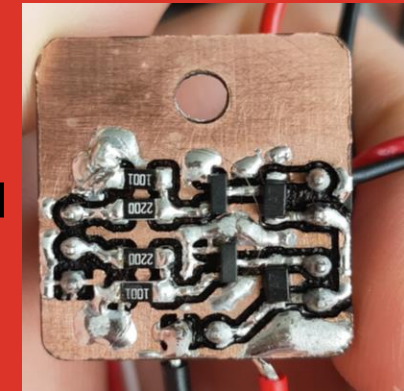
Création Kicad



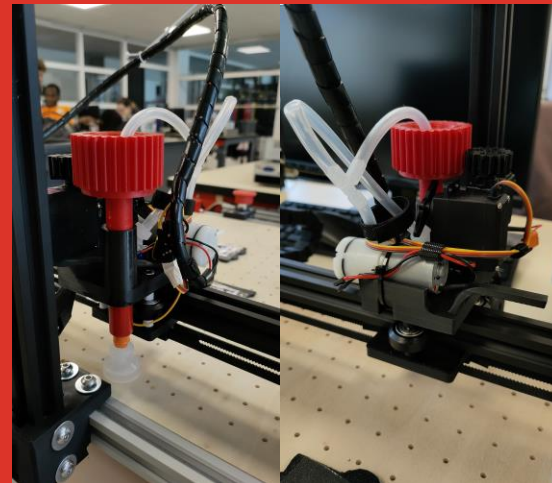
Gravure de la carte



Perçage de la carte



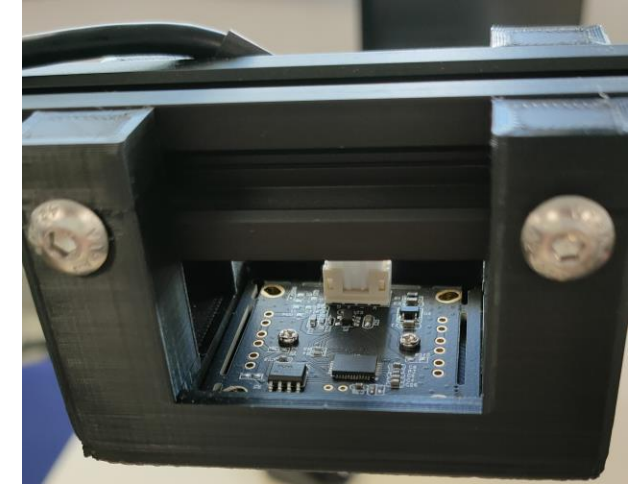
Soudage



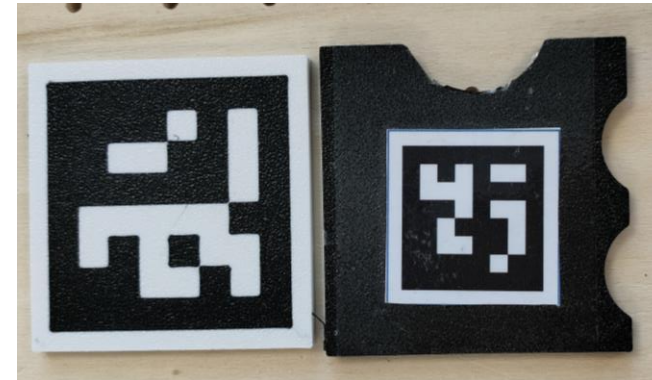
Branchement

Caméra

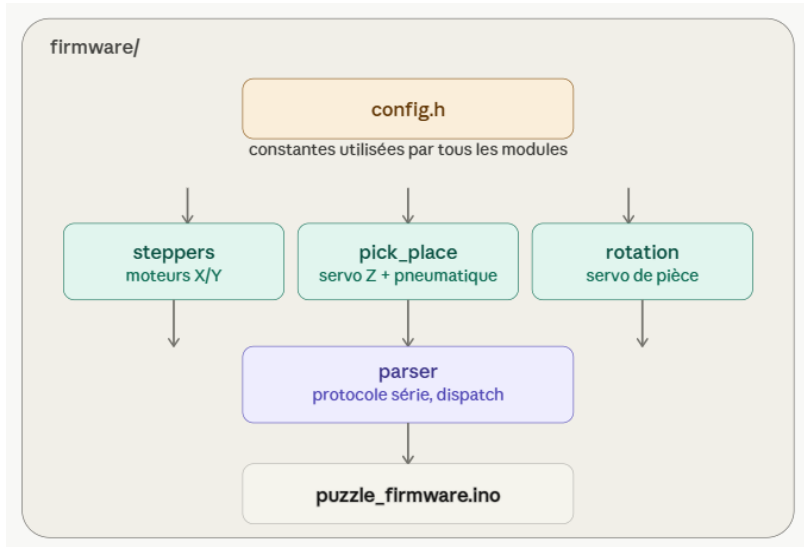
- Langage associé
- Détection des formes
- Détection des Arucos
- Précision



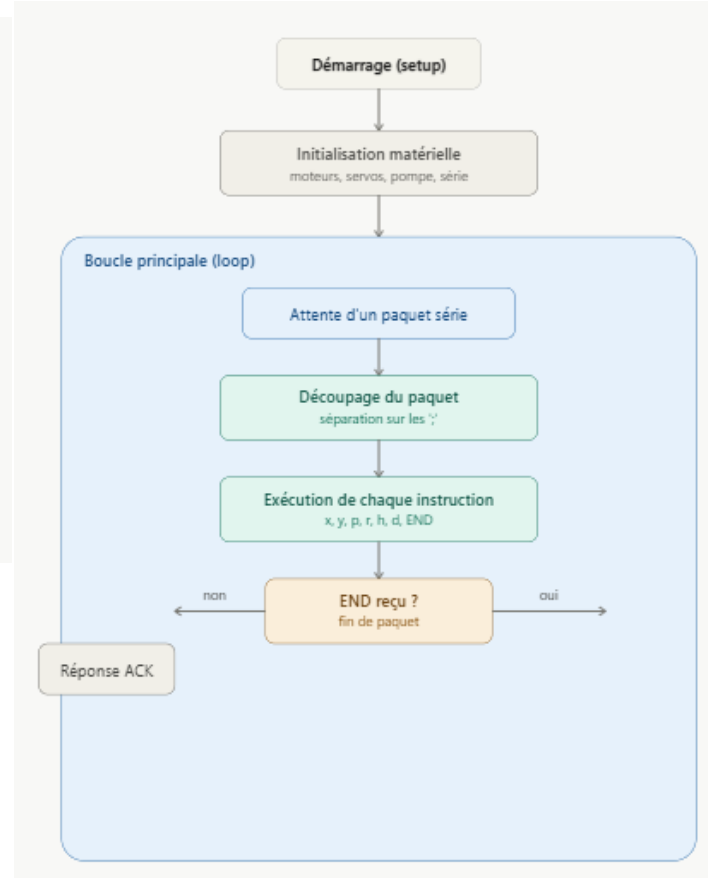
Webcam FIT0892 – DFRobot



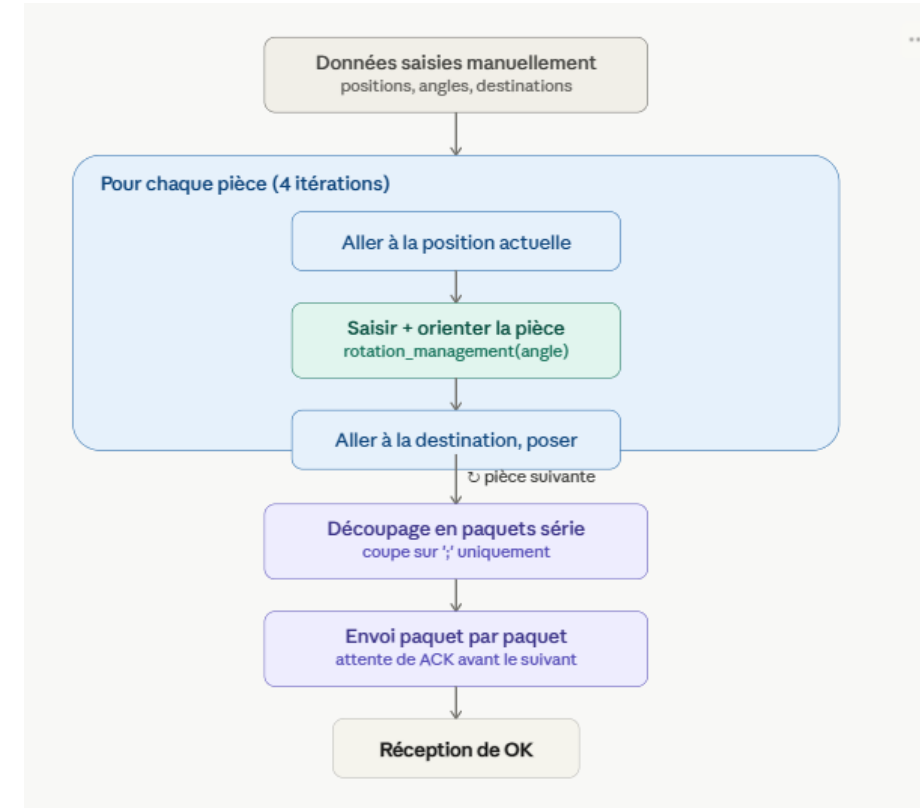
Programmation



- Folder



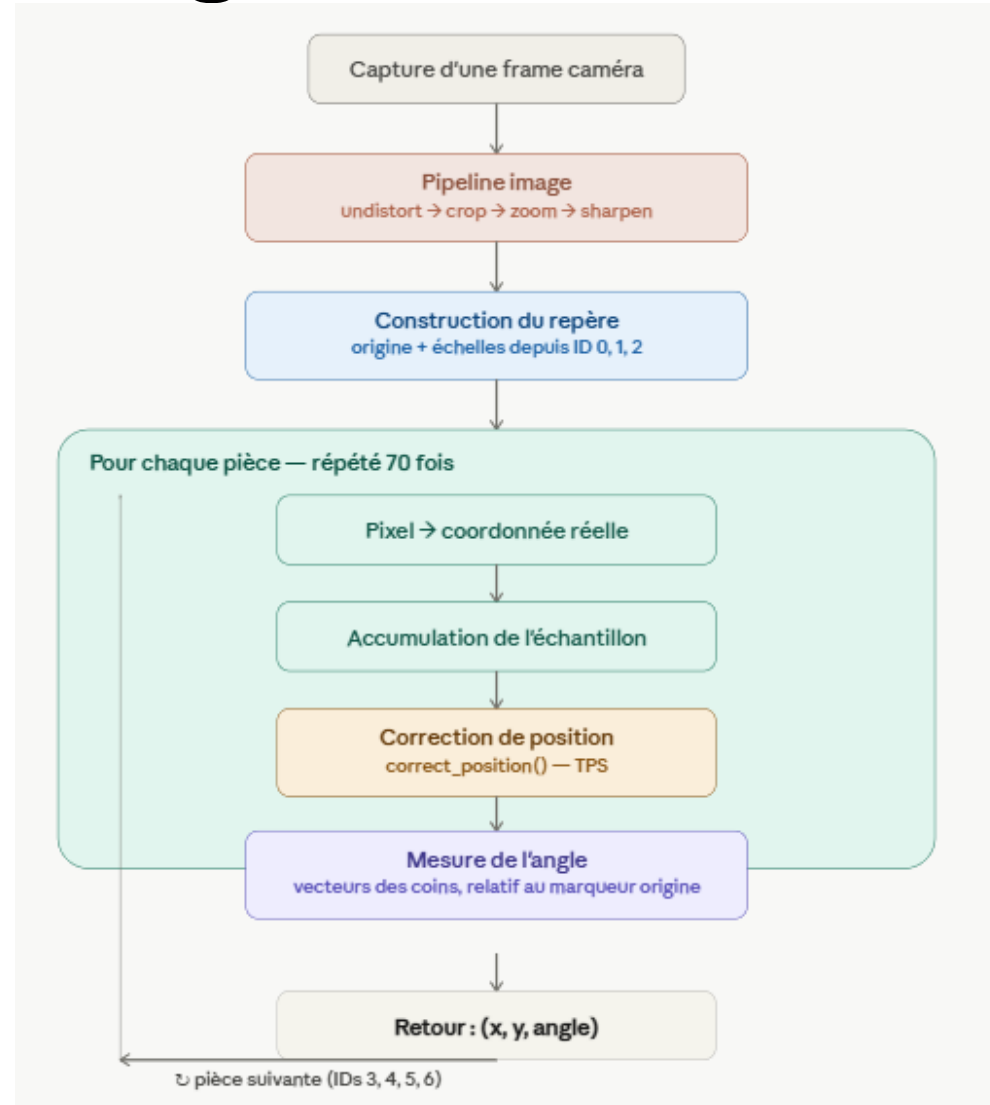
- Firmware

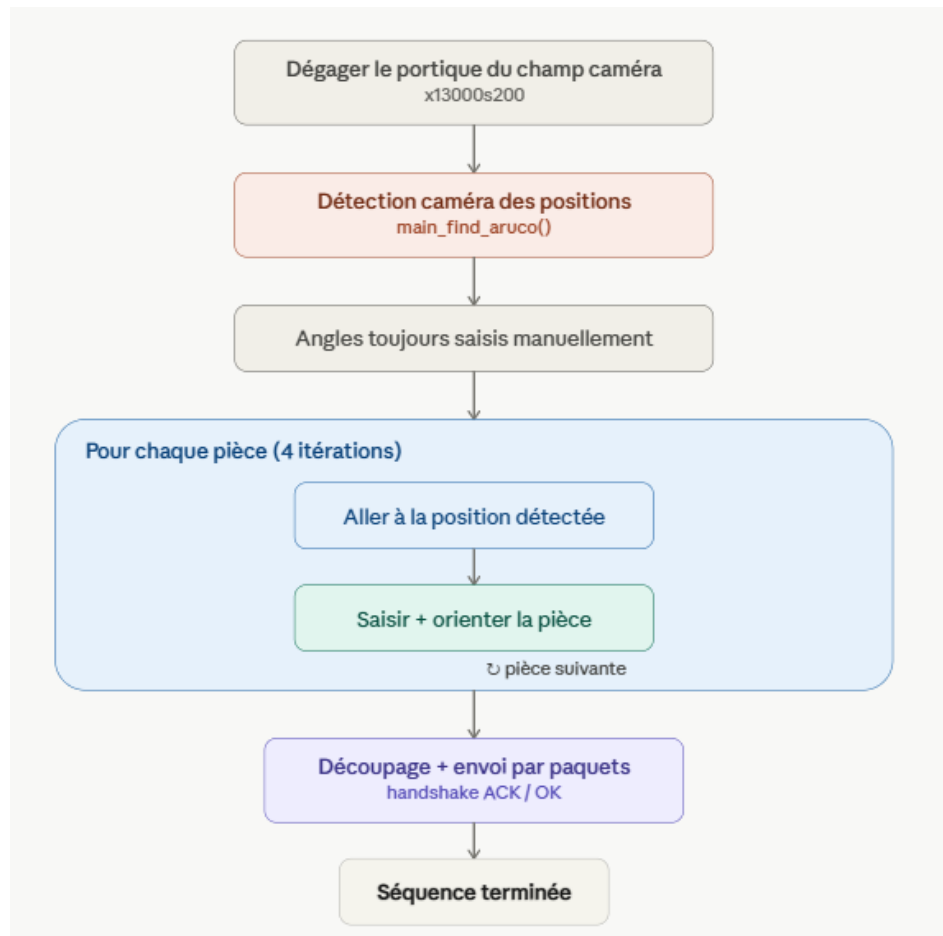


- Puzzle Solver 2

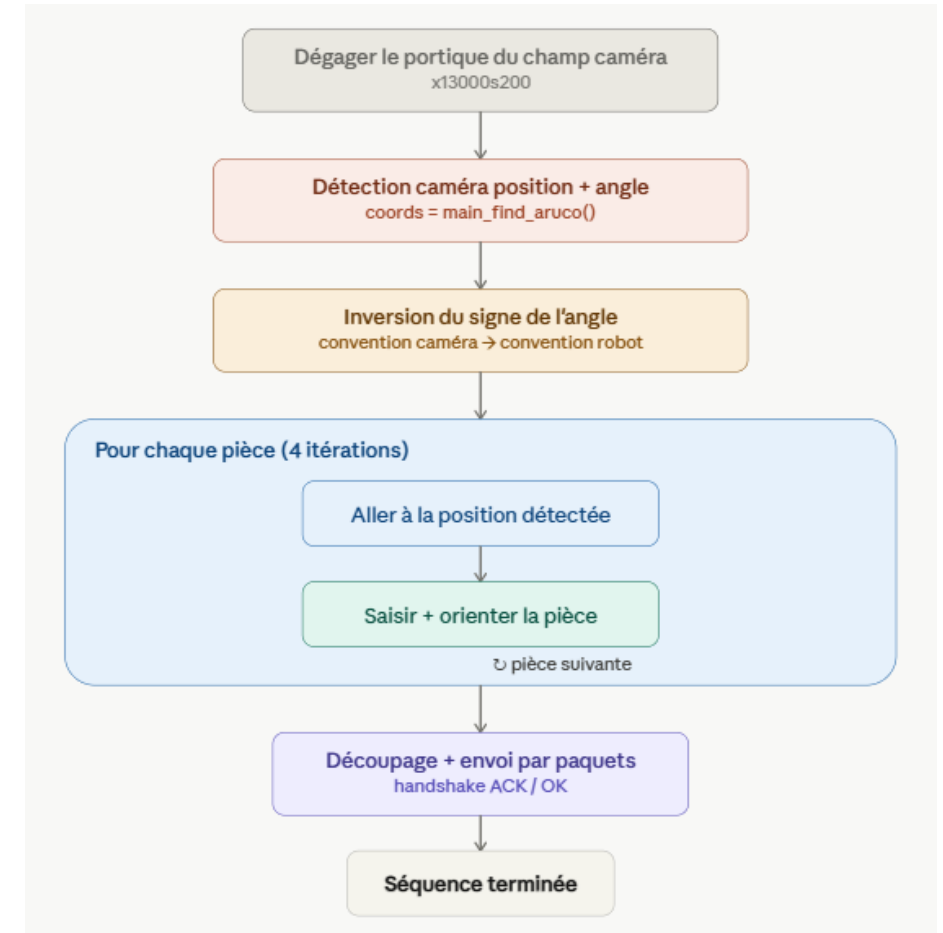
Programmation

- Détermination de la position et de l'angle de chaque pièces



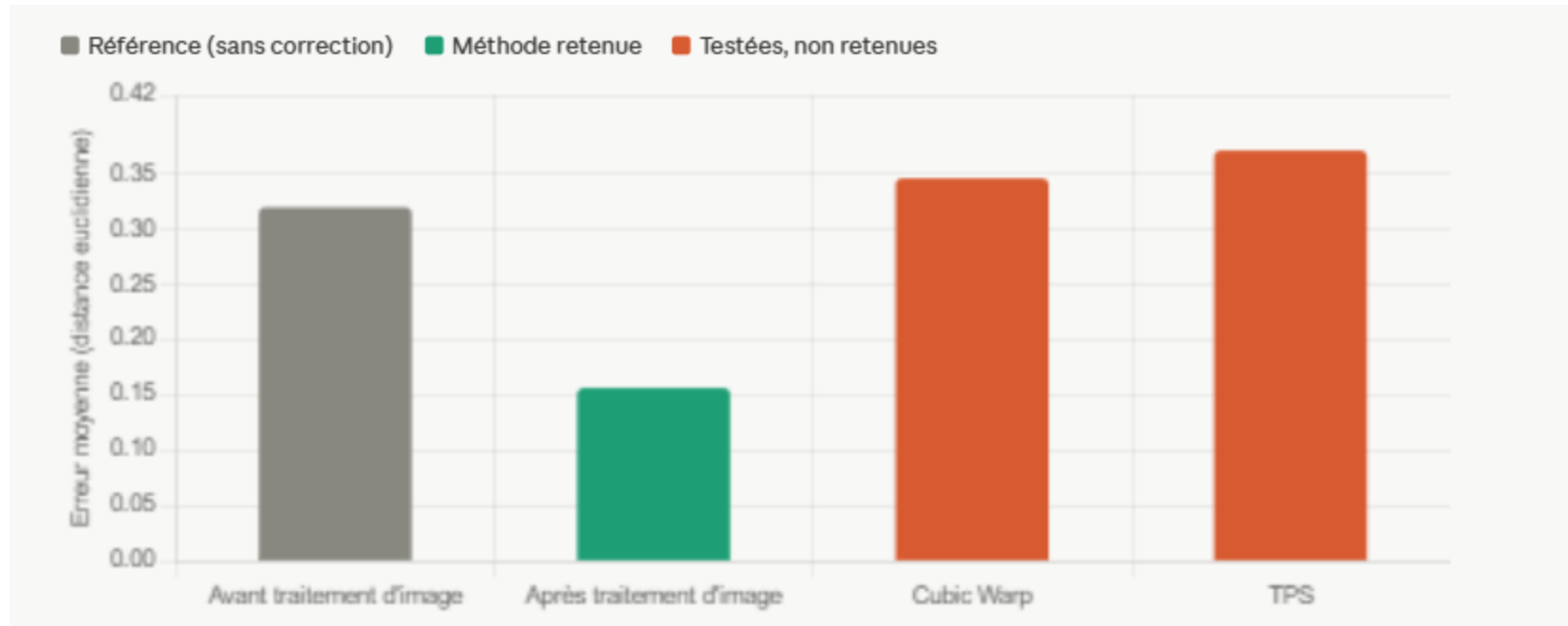


- Puzzle Solver 3

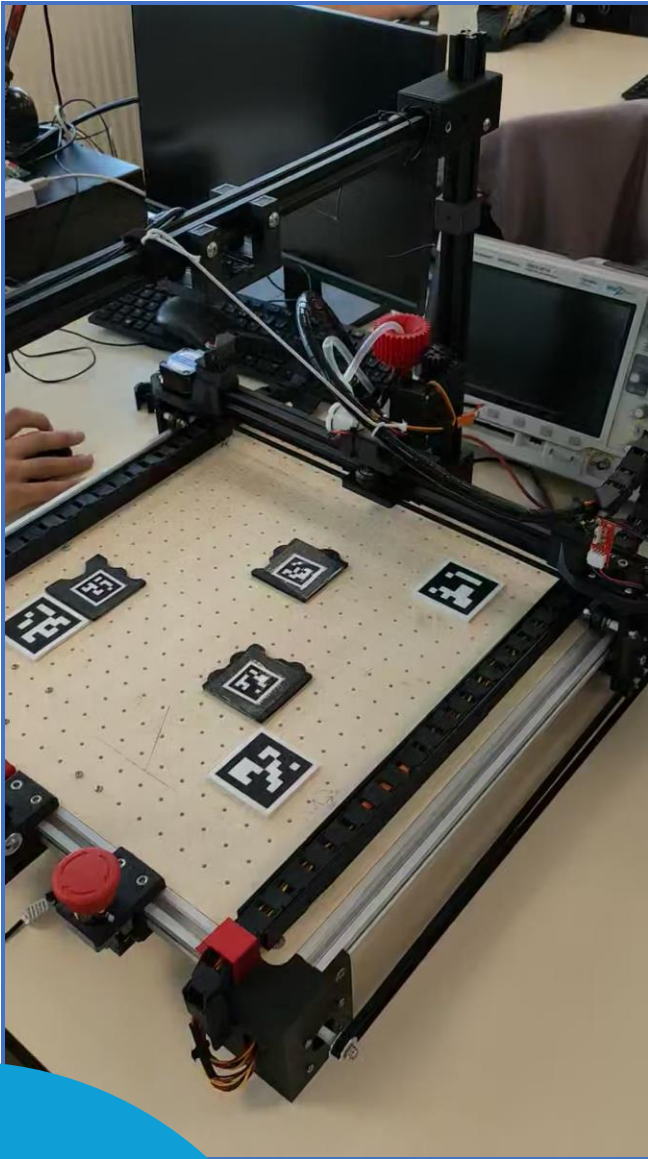


- Puzzle Solver 4

Programmation



- Algorithme de correction de position



Conclusion

- Structure mécanique fonctionnelle
- Intégration au domaine professionnel
- Possibles améliorations